

**EXAM 7 – SPRING 2014**

26. (2 points)

a. (0.5 point)

Define the following:

- i. Economic capital level
- ii. Rating agency capital

b. (0.5 point)

Discuss why a firm might calibrate the capital model for the economic capital level instead of the rating agency capital level.

c. (0.5 point)

Define the following:

- i. Economic impairment earnings
- ii. Protection of franchise value

d. (0.5 point)

Discuss why a firm might calibrate the capital model to the economic impairment earnings level instead of the protection of franchise value level.

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## SAMPLE ANSWERS AND EXAMINER'S REPORT

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| <b>QUESTION 26</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                      |
| <b>TOTAL POINT VALUE: 2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>LEARNING OBJECTIVE:</b> C8: Evaluate best practices in risk measurement, modeling, and management of various financial and non-financial risks faced by an entity |
| <b>SAMPLE ANSWERS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                      |
| <b>Part a: 0.5 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                      |
| <p><i>Sample Answer 1</i></p> <p>A1) Economic Capital Level is the amount of capital needed to keep the probability of ruin under a certain threshold<br/> A2) Rating Agency Capital is the amount of capital needed to maintain certain rating may be developed by the models from rating agency</p> <p><i>Sample Answer 2</i></p> <p>A1) Amount of Capital needed to reduce risk of insolvency below a threshold<br/> A2) Amount of capital needed to reduce risk of a rating downgrade below a certain threshold</p>                                                                                                                     |                                                                                                                                                                      |
| <b>Part b: 0.5 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                      |
| <p><i>Sample Answer 1</i></p> <p>Rating Agency Capital level is more or less dependent on the rating agency's activities. However, for a firm's economic capital level, it evaluates the risk of insolvency at a different threshold from rating agency capital level. Further, it might look at various or different types of risks.</p> <p><i>Sample Answer 2</i></p> <p>Calibrate to economic capital level to ensure solvency of the company based on the produced output. Rating Agency Capital level may not be high enough to endure a small probability of company failure</p>                                                      |                                                                                                                                                                      |
| <b>Part c: 0.5 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                      |
| <p><i>Sample Answer 1</i></p> <p>C1: Economic impairment earnings is the amount of loss that would put the firm in a state of economic impairment<br/> C2: Protection of franchise value is the amount of capital required to keep the firm's franchise value (market- book) at a tolerable level</p> <p><i>Sample Answer 2</i></p> <p>C1: Economic Impairment Earnings → the amount of losses required to put a firm in economic impairment at a given threshold<br/> C2: Protection of Franchise Value: the amount of capital required to maintain the franchise value of the firm above a certain threshold with a given probability</p> |                                                                                                                                                                      |

## SAMPLE ANSWERS AND EXAMINER'S REPORT

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| <b>Part d: 0.5 point</b>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <i>Sample Answer 1</i> <p>Firm might calibrate to EIEL instead of FVL because being impaired has more severe consequences than just losing franchise value. Company should be concerned with tail events that impact debt and policyholders not just the franchise value which mostly affects shareholders</p>                                                                                                                                       |
| <i>Sample Answer 2</i> <p>Economic impairment earnings level is often used to measure risk appetite rather than capital. The firm may be focused on measuring risk appetite as an input on decisions and maximizing its opportunities</p>                                                                                                                                                                                                            |
| <b>EXAMINER'S REPORT</b>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| This question was meant to test the candidates' recollection of definitions and their ability to compare and contrast them. The majority of candidates were able to supply the requested definitions, however, they had trouble comparing and contrasting the terms.                                                                                                                                                                                 |
| <b>Part a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <ul style="list-style-type: none"><li>• The candidates were expected to know the definitions and most candidates received full credit.</li><li>• The ECL should refer to probability of ruin or insolvency.</li><li>• Rating Agency Capital is required to achieve a certain rating; however, some candidates simply claimed it was required.</li></ul>                                                                                              |
| <b>Part b</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| There were several different possibilities for answering the question, but some candidates only listed descriptions instead of answering why. About half of the candidates received full credit.                                                                                                                                                                                                                                                     |
| <b>Part c</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| C1: Definition of Economic Impairment Earnings is a level of loss and not capital.<br>C2: The definition says we want to protect the franchise level from falling below a certain threshold. Many candidates said to protect from a decrease, or maintain the level, but they also needed to incorporate the concept of threshold<br>About half the candidates received partial credit, with the remainder split between full credit, and no credit. |
| <b>Part d</b>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Candidates referred to franchise value as difficult to value and this was not accepted. About 2/3 of the candidates received no credit on this response; most of these struggled with the definitions themselves.                                                                                                                                                                                                                                    |